

PROJECT

Production-Style Spring Boot Weather Application with Automated CI/CD Pipeline using Jenkins, Docker, Apache Tomcat, and AWS EC2 further deployed in kubernetes(killercoda).

Development part:

Launched one ec2 instance for developing the application

On that instance installed this :**Java, Javac and Maven**

```
root@ip-172-31-17-133:~# java --version
openjdk 21.0.11 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-24.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-24.04.2-Ubuntu, mixed mode, sharing)
root@ip-172-31-17-133:~# mvn --version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.11, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd64
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "6.17.0-1017-aws", arch: "amd64", family: "unix"
root@ip-172-31-17-133:~# javac --version
javac 21.0.11
root@ip-172-31-17-133:~# |
```

Uploaded starter.zip and weather.zip, extracted project in /opt directory.

Starter.zip ->

Weather.zip →

```
root@ip-172-31-17-133:/opt# wget https://start.spring.io/starter.zip \
-d dependencies=web \
-d javaVersion=21 \
-d type=maven-project \
-d groupId=com.weather \
-d artifactId=weather-app \
-o weather-app.zip
Setting --debug (debug) to 1
Setting --debug (debug) to 1
Setting --debug (debug) to 1
Setting --debug (debug) to 1
Setting --output-file (logfile) to weather-app.zip
root@ip-172-31-17-133:/opt#
root@ip-172-31-17-133:/opt# ls
starter.zip  weather-app.zip
root@ip-172-31-17-133:/opt# apt install unzip
root@ip-172-31-17-133:/opt# unzip weather-app.zip
Archive:  weather-app.zip
  inflating: mvnw.cmd
    creating: src/
    creating: src/test/
    creating: src/test/java/
    creating: src/test/java/com/
    creating: src/test/java/com/weather/
    creating: src/test/java/com/weather/weather_app/
  inflating: src/test/java/com/weather/weather_app/WeatherAppApplicationTests.java
    creating: src/main/
    creating: src/main/java/
    creating: src/main/java/com/
    creating: src/main/java/com/weather/
    creating: src/main/java/com/weather/weather_app/
  inflating: src/main/java/com/weather/weather_app/WeatherAppApplication.java
    creating: src/main/resources/
    creating: src/main/resources/templates/
    creating: src/main/resources/static/
  inflating: src/main/resources/application.properties
replace .gitattributes? [y]es, [n]o, [A]ll, [N]one, [r]ename: A
  inflating: .gitattributes
    inflating: pom.xml
    inflating: mvnw
    inflating: HELP.md
    inflating: .gitignore
    creating: .mvn/
    creating: .mvn/wrapper/
  inflating: .mvn/wrapper/maven-wrapper.properties
root@ip-172-31-17-133:/opt#
root@ip-172-31-17-133:/opt#
root@ip-172-31-17-133:/opt# ls
HELP.md  mvnw  mvnw.cmd  pom.xml  src  starter.zip  weather-app.zip
root@ip-172-31-17-133:/opt#
```

Executed ./mvnw clean package and generated WAR file.

```
root@ip-172-31-17-133:/opt#  
root@ip-172-31-17-133:/opt# ./mvnw clean package  
[INFO] Scanning for projects...
```

Build success i.e war file for this project has been generated.

```
[INFO] Replacing main artifact /opt/target/weather-app-0.0.1-SNAPSHOT.jar with repack  
aged archive, adding nested dependencies in BOOT-INF/.  
[INFO] The original artifact has been renamed to /opt/target/weather-app-0.0.1-SNAPSH  
OT.jar.original  
[INFO] -----  
[INFO] BUILD SUCCESS  
[INFO] -----  
[INFO] Total time: 14.323 s  
[INFO] Finished at: 2026-07-31T12:44:19Z  
[INFO] -----
```

Customizes index.html created front end design in index.html

```
root@ip-172-31-17-133:/opt# nano /opt/src/main/resources/static/index.html  
root@ip-172-31-17-133:/opt# ./mvnw spring-boot:run  
[INFO] Scanning for projects...  
[INFO] -----< com.weather:weather-app >-----  
[INFO] Building weather-app 0.0.1-SNAPSHOT  
[INFO] from pom.xml  
[INFO] -----[ jar ]-----  
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ weather-app >>>
```

Wrote Backend logic to fetch weather data from API

```
root@ip-172-31-17-133:/opt# nano /opt/src/main/java/com/weather/weather_app/WeatherController.java  
root@ip-172-31-17-133:/opt#
```

Created docker file for this project for containerized deployment

Pushed Customized project code to GitHub repository

```
root@ip-172-31-17-133:/opt# git init  
hint: Using 'master' as the name for the initial branch. This default branch name  
hint: is subject to change. To configure the initial branch name to use in all  
hint: of your new repositories, which will suppress this warning, call:  
hint:  
hint: git config --global init.defaultBranch <name>  
hint:  
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and  
hint: 'development'. The just-created branch can be renamed via this command:  
hint:  
hint: git branch -m <name>  
Initialized empty Git repository in /opt/.git/  
root@ip-172-31-17-133:/opt# git config --global user.name "ritupatel1104"  
root@ip-172-31-17-133:/opt# git config --global user.email "ritupatel1104@gmail.com"  
root@ip-172-31-17-133:/opt#  
root@ip-172-31-17-133:/opt#
```

Project pushed to github

```
root@ip-172-31-17-133:/opt# git add .
root@ip-172-31-17-133:/opt# git commit -m "weather app"
On branch master
nothing to commit, working tree clean
root@ip-172-31-17-133:/opt# git remote add origin https://github.com/ritupatel1104/weather-app-INDIA.git
root@ip-172-31-17-133:/opt# git branch
* master
root@ip-172-31-17-133:/opt#
root@ip-172-31-17-133:/opt#
root@ip-172-31-17-133:/opt# git push origin master
Username for 'https://github.com': ritupatel1104
Password for 'https://ritupatel1104@github.com':
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 2 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (30/30), 72.22 KiB | 10.32 MiB/s, done.
Total 30 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/ritupatel1104/Weather-app-INDIA.git
 * [new branch]      master -> master
root@ip-172-31-17-133:/opt# cd
```

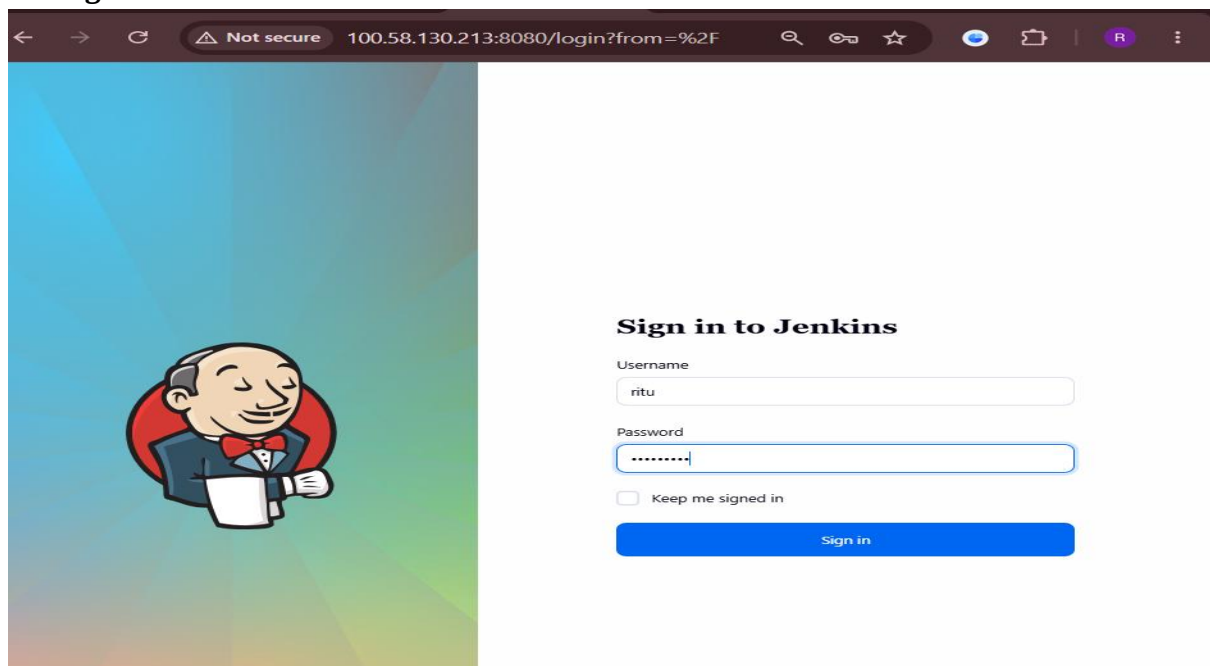
After pushing code to git terminated this instance and launched new instance

CI/CD Part

Installed Jenkins, tomcat 10, java, maven, javac and docker on that instance

```
root@ip-172-31-17-18:~# java --version
openjdk 21.0.11 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-24.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-24.04.2-Ubuntu, mixed mode, sharing)
root@ip-172-31-17-18:~# javac --version
javac 21.0.11
root@ip-172-31-17-18:~# mvn --version
Apache Maven 3.8.7
Maven home: /usr/share/maven
Java version: 21.0.11, vendor: Ubuntu, runtime: /usr/lib/jvm/java-21-openjdk-amd
Default locale: en, platform encoding: UTF-8
OS name: "linux", version: "7.0.0-1009-aws", arch: "amd64", family: "unix"
root@ip-172-31-17-18:~# jenkins --version
2.568.1
root@ip-172-31-17-18:~# docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu3~24.04.2
root@ip-172-31-17-18:~# |
```

Configured Jenkins



Creating new pipeline job for CI/CD

 **Jenkins** / New Item

🔍 ⚙️ 👤

New Item

Enter an item name

Select an item type

 **Pipeline**
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.

 **Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially.

Set trigger for automatic job build when anything is pushed to repo

Triggers

Set up automated actions that start your build based on specific events, like code changes or scheduled times.

- ☐ Build after other projects are built ?
- ☐ Build periodically ?
- ☒ GitHub hook trigger for GITScm polling ?
- ☐ Poll SCM ?
- ☐ Trigger builds remotely (e.g., from scripts) ?

Wrote pipeline script

```
1 pipeline {
2   agent any
3
4   environment {
5     APP_NAME = 'india-weather-live'
6     CONTAINER_NAME = 'india-weather-live-container'
7   }
8
9   stages {
10
11     stage('Checkout') {
12       steps {
13         git branch: 'master',
14             url: 'https://github.com/ritupatel1104/Weather-app-INDIA.git'
15       }
16     }
17
18     stage('Build WAR') {
19       steps {
20         sh 'chmod +x mvnw'
21         sh './mvnw clean package'
22       }
23     }
24
25     stage('Deploy to Tomcat') {
26       steps {
27         sh 'rm -rf /opt/tomcat/webapps/ROOT* || true'
28         sh 'cp target/*.war /opt/tomcat/webapps/ROOT.war'
29         sh '/opt/tomcat/bin/shutdown.sh || true'
30         sh 'sleep 5'
31         sh '/opt/tomcat/bin/startup.sh'
32       }
33     }
34   }
```

```

35 ▾      stage('Build Docker Image') {
36 ▾          steps {
37 ▾              sh 'docker build -t $APP_NAME .'
38 ▾          }
39 ▾      }
40
41 ▾      stage('Run Docker Container') {
42 ▾          steps {
43 ▾              sh 'docker rm -f $CONTAINER_NAME || true'
44 ▾              sh 'docker run -d --name $CONTAINER_NAME -p 8081:8080 $APP_NAME'
45 ▾          }
46 ▾      }
47
48 ▾      stage('Verify') {
49 ▾          steps {
50 ▾              sh 'docker ps'
51 ▾              sh 'ls -lh /opt/tomcat/webapps'
52 ▾          }
53 ▾      }
54 ▾  }
55
56 ▾  post {
57 ▾      success {
58 ▾          echo 'Tomcat + Docker deployment completed successfully!'
59 ▾      }
60
61 ▾      failure {
62 ▾          echo 'Pipeline failed!'
63 ▾      }
64 ▾  }
65 }

```

Executed the job



India-weather-app-deploy

Weather-app india

Permalinks

- [Last build \(#11\), 47 min ago](#)
- [Last stable build \(#11\), 47 min ago](#)
- [Last successful build \(#11\), 47 min ago](#)
- [Last failed build \(#9\), 5 hr 22 min ago](#)
- [Last unsuccessful build \(#9\), 5 hr 22 min ago](#)
- [Last completed build \(#11\), 47 min ago](#)

Pipeline overview

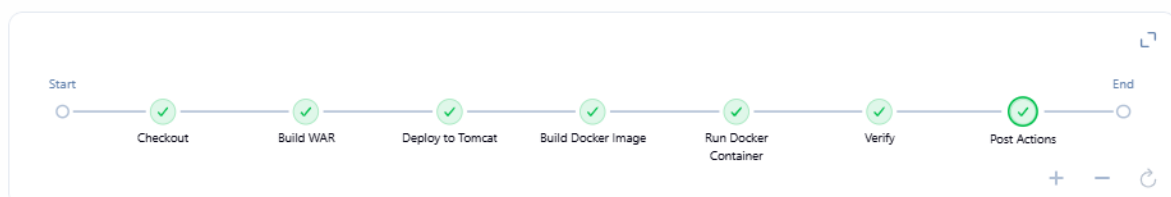


< #11

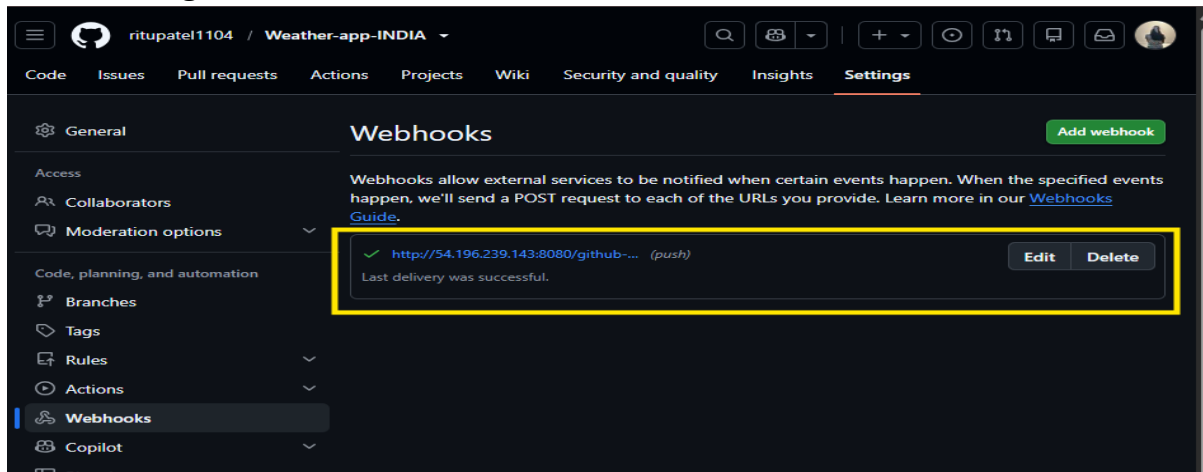
🔄 Rerun



📄 Started by GitHub push by ritupatel1104 ⌚ Started 45 min ago ⌚ Took 48 sec • </> Changes



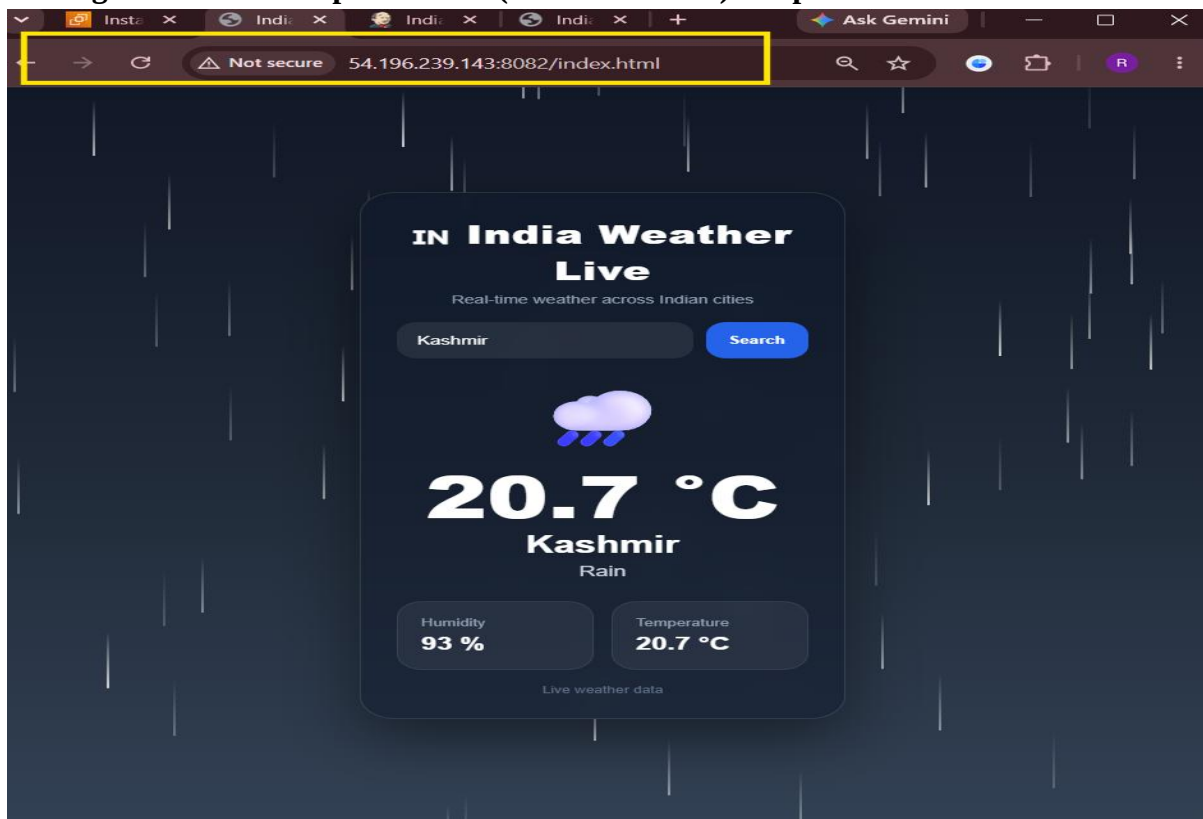
Webhook in git



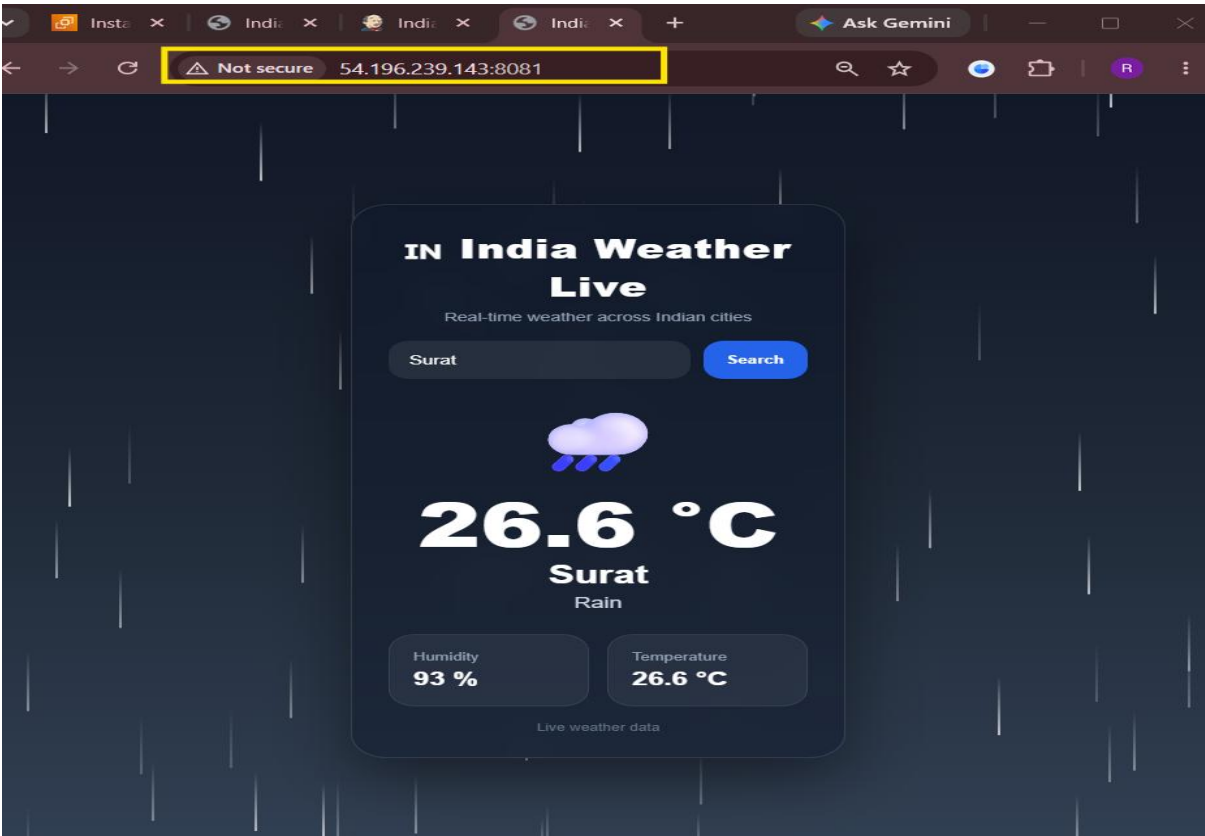
Started tomcat after pipeline executed successfully

```
root@ip-172-31-17-18:/opt# ls
apache-tomcat-10.1.48.tar.gz  apache-tomcat-9.0.120.zip  containerd  tomcat
root@ip-172-31-17-18:/opt# cd tomcat/bin
root@ip-172-31-17-18:/opt/tomcat/bin# sh startup.sh
Using CATALINA_BASE:   /opt/tomcat
Using CATALINA_HOME:   /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
root@ip-172-31-17-18:/opt/tomcat/bin#
```

Configured the tomcat port to 8082(from server file) Output



Container port to 8081



Automatic Job build due to trigger

 **Jenkins** / India-weather-app-deploy

 Stages

 Pipeline Syntax

 GitHub Hook Log

- Last unsuccessful build (#9), 5 hr 51 min ago
- Last completed build (#15), 2 min 22 sec ago

Builds >

Today

✓

#16

12:30

✓

#15

12:28

✓

#14

12:26

✓

#13

12:25

✓

#12

12:24

✓

#11

11:33

✓

#10

07:02

✗

#9

06:58

✓

#8

06:48

✗

#7

06:45

✓

#6

06:40

✓

#5

06:35

✓

#4

06:22

Image deployed to DockerHub

Name	Last Pushed ↑	Contains	Visibility	Scout
ritika1104/weather-app-india	6 days ago	IMAGE	Public	Inactive
ritika1104/shopvella-frontend	7 days ago	IMAGE	Public	Inactive
ritika1104/shopvella-backend	7 days ago	IMAGE	Public	Inactive
ritika1104/streamlit-app	10 days ago	IMAGE	Public	Inactive

Deployment on

k8s

Create deployment.yaml file and Apply deployment file

```
root@controlplane:~$ vi deploy.yaml
root@controlplane:~$ cat deploy.yaml
apiVersion: apps/v1
kind: Deployment

metadata:
  name: weather-live-deployment

spec:
  replicas: 1

  selector:
    matchLabels:
      app: weather

  template:
    metadata:
      labels:
        app: weather

    spec:
      containers:
        - name: weather
          image: ritika1104/weather-app-india:latest
          ports:
            - containerPort: 8080

root@controlplane:~$
root@controlplane:~$
root@controlplane:~$ kubectl apply -f deploy.yaml
deployment.apps/weather-live-deployment created
root@controlplane:~$
root@controlplane:~$
```

check the deployment and pods

```
root@controlplane:~$ kubectl get Deployments
NAME                    READY   UP-TO-DATE   AVAILABLE   AGE
weather-live-deployment 1/1     1            1           33s
root@controlplane:~$
root@controlplane:~$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
weather-live-deployment-74ddd85658-4pr98 1/1     Running   0           38s
root@controlplane:~$
```

Create service.yaml file and Applyservice file

```
root@controlplane:~$ vi service.yaml
root@controlplane:~$ cat service.yaml
apiVersion: v1
kind: Service

metadata:
  name: weather-live-service

spec:
  type: NodePort

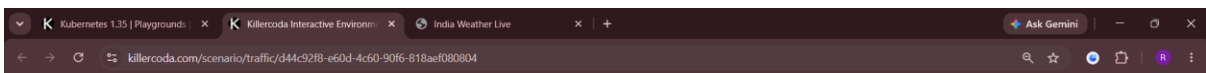
  selector:
    app: weather

  ports:
    - protocol: TCP
      port: 80
      targetPort: 8080
      nodePort: 30081
root@controlplane:~$ kubectl apply -f service.yaml
service/weather-live-service created
root@controlplane:~$
```

Check the service created

```
root@controlplane:~$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
kubernetes           ClusterIP   10.96.0.1     <none>         443/TCP          17d
weather-live-service NodePort    10.104.202.51 <none>         80:30081/TCP     11s
root@controlplane:~$
```

Access the application from no 30081 port in killercods's traffic port



Traffic Port Accessor

Access HTTP services which run in your environment

The services need to run on all interfaces (like 0.0.0.0) and not just localhost

Host 1

Common Ports

80 8080

Custom Ports

30081

Access

Host 2

Common Ports

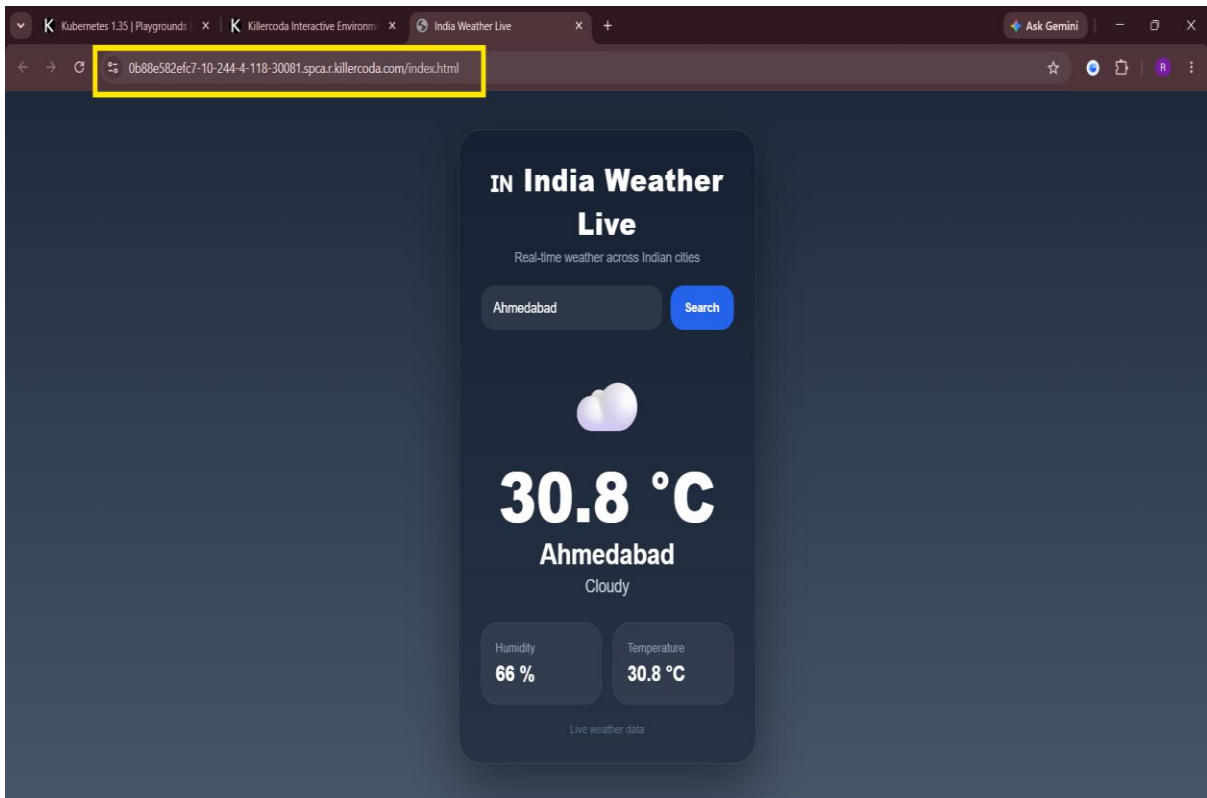
80 8080

Custom Ports

1234

Access

Application Deployed Successfully on Killercode



Source Code for the application: <https://github.com/ritupatel1104/Weather-app-INDIA>